

$$\int \frac{\sqrt[6]{e^2-1}}{\sqrt[6]{e-1}} \frac{1}{\left(\frac{1}{x^5}+x\right)\ln(1+x^6)}dx$$

$$\int \frac{\sqrt[6]{e^2-1}}{\sqrt[6]{e-1}} \frac{x^5}{(1+x^6)\ln(1+x^6)}dx$$

$$u:=\ln(1+x^6)$$

$$du=6\frac{x^5}{1+x^6}dx$$

$$\frac{1}{6}\int_1^2\frac{1}{u}du$$

$$\frac{1}{6}\ln|u|\Big|_1^2$$

$$\frac{\ln 2}{6}$$